

Data Flow Diagram

Date	26 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Data Flow Diagram

Data Flow Diagram (DFD) illustrating how soil nutrient and environmental data flow through the OptiCrop: Smart Agricultural Production Optimization Engine. It represents the interaction between the farmer, the machine learning prediction system, and the crop dataset to generate intelligent crop recommendations based on soil and climatic conditions.

Symbol	Name	Description
Oval	External Entity	Represents the Farmer or Agricultural User who interacts with the OptiCrop system by providing soil and environmental details.
Numbered Rectangle	Process	Represents the machine learning processes such as receiving input parameters, analyzing data, and predicting the most suitable crop.
Solid Rectangle	Data Store	Stores the crop dataset, trained machine learning model, and prediction-related information used by the system.
Arrow	Data Flow	Represents the flow of information between the farmer, machine learning model, dataset, and the recommendation system.

DFD - OptiCrop: Smart Agricultural Production Optimization Engine

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